LEGEND

Strong Entity | Weak Entity

Relationship | Identifying Rel.

Attribute | Key Attribute

Derived Attr. | Multi-value Attr.

>= Double line = Total Participation
1:N / M:N = Cardinality labels

ALL ENTITIES (14 Sections):
1. STATION (strong)
2. PLATFORM (weak of STATION)
3. LINE (strong)
4. STATION_ON_LINE (M:N rel w/ attrs)
5. TRAIN (strong)
6. TRAIN_SCHEDULE (strong)
7. TRAIN_STOP (weak of SCHEDULE)
8. TRACK (strong)
9. DELAY_EVENT (strong)
10. ALERT (strong)
11. LIVE_TRACKING (strong)
12. RESCHEDULE_LOG (strong)
13. MAINTENANCE_LOG (strong)
14. TRAIN_INCIDENT (strong)
15. DRIVER (strong)
16. DRIVER_SHIFT (weak of DRIVER)
17. PASSENGER (strong)
18. TICKET (strong)
19. PAYMENT (strong)
20. TRAVEL_PASS (strong)
21. FARE_RULE (strong)
22. EMPLOYEE (strong)
23. MAINTENANCE_LOG (strong)
24. FEEDBACK (strong)

★ SECTION 1: INFRASTRUCTURE — Station, Platform, Line, Station-on-Line

★ SECTION 2: OPERATIONS — Train, Train Schedule, Train Stop

★ SECTION 3: DELAY, ALERT & LIVE TRACKING

★ SECTION 4: DRIVER & SHIFTS

★ SECTION 5: PASSENGER, TICKET, PAYMENT

★ SECTION 6: FARES & TRAVEL PASS

★ SECTION 7: EMPLOYEE & MAINTENANCE

★ SECTION 8: TRACK

★ SECTION 9: RESCHEDULE LOG

★ SECTION 10: STATION CLOSURE (Temporary Removal)

★ SECTION 11: TRAIN INCIDENT (Crash / Breakdown)

★ SECTION 12: PASSENGER TRAVEL LOG — Travelling In (determines crowd)

★ SECTION 13: TICKET ↔ TRAIN_STOP (QR Entry/Exit validation)

★ SECTION 14: PASSENGER FEEDBACK

sequence_no
is_active
dist_from_start_km
STATION_ON_LINE
total_length_km
line_id
color_code
line_name
operational_since
status
LINE
N
M
1
zone_no
location
is_active
latitude
longitude
station_name
station_id
opening_date
station_type
STATION
HAS_PLATFORM
N
direction
capacity
PLATFORM
platform_no
is_available
RUNS_ON
TRACKED_BY
SCHEDULED_ON
tracking_id
speed_kmph
dynamic_block_dist_m
recorded_at
LIVE_TRACKING
updated_ETA
longitude
latitude
full_name
email
passenger_id
phone
registered_at
date_of_birth
PASSENGER
passenger_type
HOLDS_PASS
disability_flag
BOOKS
pass_id
pass_type
valid_from
valid_to
institution_name
price
TRAVEL_PASS
loss_id_covered
manufacturer
train_id
operational_status
train_type
train_number
current_location
current_speed_kmph
commission_date
total_capacity
current_crowd_count
is_on_rest
TRAIN
RUNS
schedule_id
departure_time
arrival_time
day_type
TRAIN_SCHEDULE
assigned_track_no
direction
schedule_status
normal_ETA
CAUSES_DELAY
delay_id
delay_minutes
reason
reported_at
affected_station_id
DELAY_EVENT
resolution_status
tp_code
ticket_id
ticket_type
issued_at
price_paid
from_station_id
to_station_id
status
booking_channel
TICKET
PRICED_BY
max_distance_km
fare_rule_id
max_distance_km
base_fare
normal_fare
senior_fare
disabled_fare
child_fare
student_fare
effective_from
FARE_RULE
STOPS_AT
HAS_STOP
stop_sequence
scheduled_arrival
halt_duration_min
scheduled_departure
TRAIN_STOP
is_active
platform_no
dynamic_ETA
PAID_VIA
payment_method
amount
payment_id
transaction_ref
paid_at
status
PAYMENT
alert_id
alert_type
message
severity
is_resolved
created_at
ALERT
GENERATES
HAS_TRACK
direction
track_id
ASSIGNED_TO_TRACK
TRACK
length_km
track_number
is_active
track_status
UNDERGOES
cost
description
maintenance_date
status
lpg_id
maintenance_type
MAINTENANCE_LOG
PERFORMS
reason
original_dep_time
reschedule_id
logged_at
new_dep_time
old_track_no
done_by_emp_id
new_track_no
RESCHEDULE_LOG
HAS_RESCHEDULE
TRIGGERS
closure_type
start_datetime
closure_id
end_datetime
status
STATION_CLOSURE
approved_by_emp
reason
HAS_CLOSURE
incident_datetime
incident_type
location_desc
status
new_train_added_id
incident_id
severity_level
reported_by_emp
TRAIN_INCIDENT
INVOLVED_IN
feedback_id
rating_1to5
comment
category
status
submitted_at
FEEDBACK
GIVES
FEEDBACK_FOR
full_name
experience_years
joining_date
employment_status
salary
phone
date_of_birth
driver_id
license_no
DRIVER
DRIVES
assignment_date
ASSIGNED_SHIFT
shift_id
shift_end
shift_date
working_hours
shift_start
DRIVER_SHIFT
MANAGES
employee_id
full_name
joining_date
role
access_level
email
department
salary
EMPLOYEE
TRAVELLING_IN
exited_at
boarded_at
status
FOR_JOURNEY
scan_timestamp
stop_role